Longitudinal Changes in Quadriceps Morphology over the First 3 Months after Anterior Cruciate Ligament Reconstruction

CHRISTOPHER D. JOHNSTON[1], DEREK R. DEWIG[2], BRIAN PIETROSIMONE[3], DARIN PADUA[3], ERIC D. RYAN[3], JOE HART[4], JEFFREY SPANG[4], and TROY BLACKBURN[3]

[1]Magnet School Assistance Program, Beaufort County School District, Beaufort, SC; [2]Department of Health and Human Performance, Fairmont State University, Fairmont, WV; [3]Department of Exercise and Sport Science, The University of North Carolina at Chapel Hill, Chapel Hill, NC; and [4]Department of Orthopaedics, School of Medicine, The University of North Carolina at Chapel Hill, Chapel Hill, NC

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ABSTRACT

JOHNSTON, C. D., D. R. DEWIG, B. PIETROSIMONE, D. PADUA, E. D. RYAN, J. HART, J. SPANG, and T. BLACKBURN. Longitudinal Changes in Quadriceps Morphology over the First 3 Months after Anterior Cruciate Ligament Reconstruction. *Med. Sci. Sports Exerc.*, Vol. 56, No. 5, pp. 933–941, 2024. **Purpose:** Neuromuscular deficits and atrophy after anterior cruciate ligament reconstruction (ACLR) may be accompanied by changes in muscle composition and poor quadriceps muscle quality (QMQ). Quadriceps atrophy occurs after ACLR but improves within the first three postoperative months, yet this hypertrophy could be attributable to increases in noncontractile tissue (i.e., poor QMQ). The purposes of this study were to evaluate changes in QMQ after ACLR and to determine if changes in QMQ and cross-sectional area (CSA) occur in parallel or independently. **Methods:** A longitudinal prospective cohort design was implemented to evaluate QMQ and CSA in 20 individuals with ACLR and 12 healthy controls. Participants completed three testing sessions (baseline/presurgery, 1 month, and 3 months) during which ultrasound images were obtained from the vastus lateralis (VL) and rectus femoris (RF). QMQ was calculated as the echo intensity (EI) of each image, with high EI representing poorer QMQ. Anatomical CSA was also obtained from each image. **Results:** RF and VL EI were greater at 1 and 3 months in the ACLR limb compared with baseline and the contralateral limb and did not change between 1 and 3 months. VL and RF CSA in the ACLR limb were smaller at 1 and 3 months compared with the contralateral limb and controls (VL only) but increased from 1 to 3 months. Changes in QMQ and CSA were not correlated. **Conclusions:** QMQ declines within the first month after ACLR and does not improve by 3 months although hypertrophy occurs, suggesting that these morphological characteristics change independently after ACLR. Poorer QMQ represents greater concentration of noncontractile tissues within the muscle and potentially contributes to chronic quadriceps dysfunction observed after ACLR. **Key Words:** MUSCLE QUALITY, DYSFUNCTION, REHABILITATION, ATROPHY, HYPERTROPHY

Anterior cruciate ligament reconstruction (ACLR) incurs a substantial risk of long-term joint health complications, with up to 80% of patients developing posttraumatic knee osteoarthritis within 10–15 yr (1–4). Chronic quadriceps dysfunction (i.e., declines in strength and voluntary activation capacity) is highly common after ACLR and contributes to aberrant gait biomechanics linked to knee osteoarthritis development (5–9). Understanding mechanisms that contribute to quadriceps dysfunction is likely critical for improving rehabilitation and enhancing long-term joint health after ACLR.

Quadriceps dysfunction after ACLR is partly attributable to altered neural factors that reduce voluntarily activation capacity (5,10). Morphological changes that impair force-producing capacity such as atrophy and preferential loss of Type II muscle fibers have also been observed after ACLR (11–15). An increase in fibrogenic cell density has also been reported in the quadriceps after ACL injury, leading to a greater concentration of fibrous, noncontractile tissue (13,16,17). Poorer quadriceps muscle quality (QMQ) is attributable to a greater concentration of noncontractile (e.g., fatty and fibrous tissue) elements within the muscle (18) caused by decreases in activity, muscle activation deficits, mobility limitations, and injury (19). Poor QMQ is associated with impaired strength and functional ability in the elderly (20–22). Additionally, poorer QMQ has been reported in the affected limb in individuals with knee osteoarthritis compared with the contralateral limb and healthy controls (23–27) and is associated with worsening symptoms (28) and poorer function (29,30) in these individuals. Similarly, poorer QMQ has been reported in the ACLR limb compared with the contralateral limb (31,32) and is associated with poorer quadriceps strength, function, and quality of life in individuals with ACLR (32–35).

Address for correspondence: Troy Blackburn, Ph.D., A.T.C., The University of North Carolina at Chapel Hill, 124 Fetzer Hall, CB # 8700, Chapel Hill, NC 27599-8700; E-mail: troyb@email.unc.edu.
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Biopsy is the gold standard for evaluating muscle composition, but this invasive and potentially painful procedure is not feasible in many clinical settings. QMQ can be assessed via MRI and CT imaging (31,36), but these approaches are time and cost prohibitive for longitudinal assessment in the clinical setting. Conversely, ultrasound is a more cost-effective imaging modality that can be used by a wide range of clinical providers to obtain a surrogate measure of muscle quality (echo intensity [EI]) (37,38). EI quantifies the gray scale brightness of an ultrasound image and is highly correlated with intramuscular fat concentration derived from MRI (18,39) and provides a valid estimate of intramuscular fat content in individuals with ACLR (40).

Declines in QMQ have been reported as early as 12 months (31) and as late as ~4 yr (32) post-ACLR, but the time course of these changes is unknown. Understanding when QMQ begins to decline after ACLR may reveal mechanistic insight regarding chronic quadriceps dysfunction, identify rehabilitation targets, and provide clinicians a tool to track progression throughout the rehabilitation process. Additionally, QMQ is associated with strength (22,34) and may serve as a proxy for quadriceps function early after ACLR when maximal effort strength testing is not feasible because of the need to limit graft loading.

The primary purpose of this study was to evaluate changes in QMQ early after ACLR. We hypothesized that QMQ would decline over the first 3 months post-ACLR compared with the contralateral limb and healthy controls. Quadriceps cross-sectional area (CSA) declines rapidly after ACLR but begins to increase within the first three to four postoperative months (11,41). However, this hypertrophy could be attributable in part to an increase in noncontractile rather than contractile tissue (i.e., poor QMQ), potentially masking underlying morphological changes. Therefore, a secondary purpose of this study was to determine if changes in QMQ and CSA occur in parallel or independently. We hypothesized that parallel changes would be observed in these morphological characteristics over the first 3 months post-ACLR.

METHODS

Experimental design. A prospective, longitudinal cohort design was used to evaluate changes in QMQ between individuals with ACLR and uninjured controls over the first 6 months post-ACLR. However, the COVID-19 pandemic interrupted longitudinal data collection such that only one participant completed the 6-month assessment. As such, the data reported here only include individuals who completed each testing session before the 6-month time point (baseline/presurgical, 1 month, 3 months). All participants completed three testing visits over a 3-month time period. For the ACLR group, baseline testing was completed after a presurgical clinic visit (mean ± SD, 14 ± 8 d before ACLR), and subsequent sessions aligned with follow-up clinic visits at 1 and 3 months post-ACLR. The control group's initial session served as a baseline, and follow-up visits were conducted 1 and 3 months after the baseline session.

Subjects. Few studies have reported QMQ assessed via ultrasound EI after ACLR, thus limiting our ability to conduct *a priori* power analysis. Garcia et al. (32) reported greater EI in the vastus lateralis (VL) of the ACLR limb compared with the contralateral limb an average of 51 ± 29 months post-ACLR (Cohen's $d = 1.50$), indicating that a sample of six individuals with ACLR would provide power of 0.80 to detect a difference in EI between limbs. Lindström et al. (31) reported poorer QMQ in the ACLR limb compared with the contralateral limb 12 months post-ACLR using CT imaging (Cohen's $d = 0.77$), indicating that a sample of 16 individuals with ACLR would provide power of 0.80 to detect a difference between limbs.

To account for differences in time since ACLR and assessment methods relative to the aforementioned studies (31,32), we recruited 30 individuals with ACLR and 24 healthy controls who volunteered to participate in the study. Of this sample, 32 participants (20 ACLR and 12 healthy controls) completed testing at the baseline, 1-month, and 3-month time points. The study was approved by an Institutional Review Board, and all participants provided written informed consent before participation. Participants in the ACLR group were recruited from a local orthopedics clinic if they had sustained primary, unilateral ACL injury and were scheduled for ACLR. Participants were excluded if they had a history of other lower extremity surgery, previous ACL injury to either limb, concussion or a lower extremity injury (other than primary ACL injury) within the 6 months before participation, or neurological disorder. Healthy controls were recruited if they had no history of lower extremity surgery, lower extremity musculoskeletal injury or concussion within the previous 6 months, or any neurological condition.

Assessments. Bilateral ultrasound images of the VL and rectus femoris (RF) were obtained at 50% of the distance between the greater trochanter and the lateral aspect of the femoral condyle (42). A bolster was placed under the test knee, as many participants could not achieve full extension at baseline and 1 month post-ACLR. Panoramic B-mode ultrasound images (gain = 56 dB, depth = 6 cm, frequency = 10 MHz) were obtained using a multifrequency linear array probe (LOGIQ e5; General Electric Company, Milwaukee, WI). Transmission gel was liberally applied to the skin to enhance acoustic coupling and minimize image artifact. A high-density foam pad was positioned perpendicular to the longitudinal axis of the thigh to stabilize the probe as it was moved lateral to medial in the transverse plane with consistent speed and minimal pressure to minimize motion artifact and to avoid compression of the underlying structures.

Research assistants blinded to limb (involved vs uninvolved) used ImageJ software (National Institutes of Health, Bethesda, MD) for all ultrasound image analyses. Each image was first scaled from pixels to cm. The region of interest (ROI) was then selected, including as much of the muscle as possible and excluding the hyperechoic fascia (Fig. 1). Anatomical CSA was calculated as the area (cm^2) of the ROI. EI was assessed using a computer-aided gray scale, whereby a value between 0 (Black) and 255 (White) arbitrary units (AU) was assigned to each pixel

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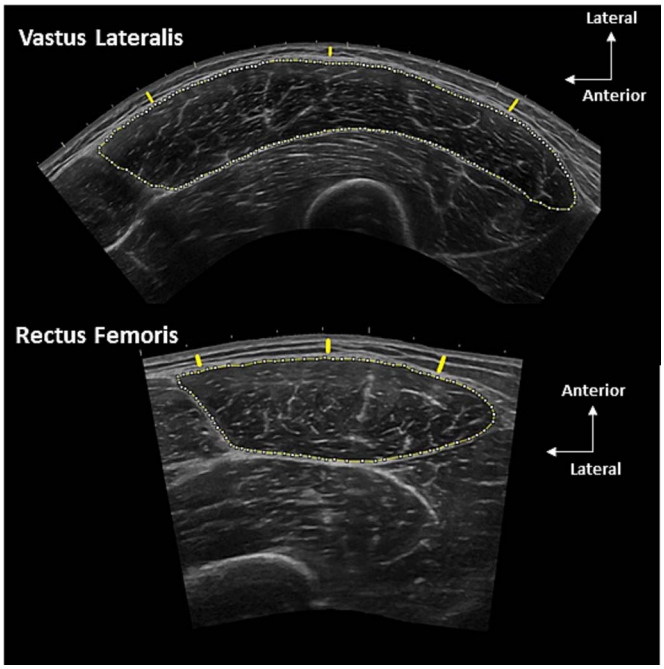

FIGURE 1—Panoramic ultrasound images of the VL (*top*) and RF (*bottom*). An ROI was selected for each muscle for which CSA and gray scale EI were calculated. Echo intensity values were corrected for subcutaneous fat thickness assessed at three locations over each muscle (*yellow lines*).

from which the mean across the ROI was calculated such that higher EI values represented greater noncontractile tissue concentration (i.e., fat, fibrous tissue) and poorer QMQ (18,39,43). Ultrasound signals are attenuated as they travel through subcutaneous fat, resulting in a darker image at greater depths; thus, EI values were corrected for subcutaneous fat thickness [corrected EI = raw EI + (SFT × 40.5278)] (39,43,44). Subcutaneous fat thickness was calculated as the straight-line distance between the deep skin border and the superficial fascia at three locations (lateral, middle, and medial) overlying the muscle of interest (Fig. 1), and the mean of the three distances was used for EI correction.

Statistical analyses. All outcomes were confirmed as being normally distributed using the Shapiro–Wilk test and visual inspection of histograms. Demographic characteristics were compared between the groups via independent-samples *t*-test for continuous variables and χ^2 analysis for categorical variables. EI and CSA in the involved (ACLR) limb were compared with healthy controls via 3 (time) × 2 (group) mixed-model repeated-measures ANOVA ($\alpha = 0.05$). The right limb was selected as the involved limb for controls. Significant interaction effects were evaluated *post hoc* via between-group comparisons at each time point and across time within each group using Bonferroni-adjusted *t*-tests ($\alpha = 0.05/9 = 0.006$ per comparison). Interlimb comparisons of EI and CSA in the ACLR group were evaluated via 3 (time) × 2 (limb) repeated-measures ANOVA ($\alpha = 0.05$). Significant interaction effects were evaluated *post hoc* via between-limb comparisons at each time point using Bonferroni-adjusted *t*-tests ($\alpha = 0.05/3 = 0.017$ per comparison).

As a secondary analysis, we also calculated change scores for each outcome over successive time points (e.g., 100 × [(1 month–baseline)/baseline]) and the associated 95% confidence intervals separately for each group/limb. Change scores for which the 95% confidence interval did not include 0 were considered statistically significant. We also used simple Pearson correlations to evaluate associations between the change scores for EI and CSA between successive time points (i.e., 1 month–baseline and 3 months–1 month) exclusively in the ACLR cohort.

The minimal clinically important differences (MCID) for EI and CSA were estimated for changes from baseline to 1 and 3 months, respectively, in the control group. MCID were defined as 1× the SEM calculated from change scores (i.e., 1 month–baseline and 3 months–baseline) as SEM = SD* $\sqrt{1-r}$, where SD is the standard deviation and *r* is the reliability coefficient (Cronbach's α). These values were then used to evaluate the clinical relevance of changes noted in the ACLR group.

RESULTS

Descriptive statistics are presented in Table 1. No demographic differences were present between the groups. Change scores for each outcome are provided in Table 2.

QMQ (EI). The time x group interaction effects were significant for the EI of the RF ($P = 0.004$, Fig. 2) and VL ($P < 0.001$, Fig. 3). *Post hoc* analyses indicated that the EI values of both RF and VL in the ACLR limb were significantly greater at 1 month ($P = 0.001$ and $P < 0.001$, respectively) and 3 months ($P = 0.001$ and $P < 0.001$, respectively) compared with baseline but did not differ between 1 and 3 months ($P = 0.826$ and 0.965, respectively). There were no differences in EI across time in the control group for either muscle ($P = 0.012$–0.885).

The EI of the RF did not differ between the ACLR and control groups at baseline ($P = 0.937$), 1 month ($P = 0.313$), or 3 months ($P = 0.232$). Similarly, the EI of the VL did not differ between the groups at baseline ($P = 0.507$), 1 month ($P = 0.079$), or 3 months ($P = 0.043$). However, the MCID values for EI of the RF and VL were 2.16 and 1.86 AU, respectively, at 1 month and 2.21 and 1.78 AU, respectively, at 3 months. At both 1 and 3 months, the difference in the EI of the RF and VL between the ACLR and the control groups exceeded the MCID.

TABLE 1. Subject demographics (mean ± SD).

	ACLR (*n* = 20)	Control (*n* = 12)	*P*
Age (yr)	21 ± 6	22 ± 3	0.884
Height (m)	1.76 ± 0.10	1.74 ± 0.08	0.575
Mass (kg)	84.8 ± 18.6	76.2 ± 14.7	0.183
Sex	15 men, 5 women	8 men, 4 women	0.612
Baseline (days pre-ACLR)	14 ± 8	NA	
1 month (days post-baseline)	33 ± 5	32 ± 6	0.687
3 months (days post-baseline)	92 ± 10	88 ± 4	0.158
Graft type	13 patellar tendon, 5 quadriceps tendon, 1 hamstring tendon, 1 allograft	NA	

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TABLE 2. Change scores.

	RF EI	VL EI	RF CSA	VL CSA
ACLR				
Baseline vs 1 month	**9.83 (4.68 to 14.98)**	**11.45 (7.70 to 15.19)**	**–15.09 (–23.12 to –7.06)**	**–25.06 (–31.50 to –18.62)**
Baseline vs 3 months	**9.40 (4.19 to 14.62**	**11.50 (7.71 to 15.29)**	–5.22 (–12.85 to 2.40)	**–12.48 (–18.96 to –6.01)**
1 month vs 3 months	–0.18 (–3.32 to 2,96)	0.27 (–3.08 to 3.62)	**13.36 (5.36 to 21.37)**	**17.85 (11.74 to 23.95)**
Contralateral				
Baseline vs 1 month	0.09 (–3.97 to 4.14)	3.66 (–0.15 to 7.46)	–0.09 (–11.47 to 11.29)	–4.21 (–9.50 to 1.07)
Baseline vs 3 months	0.35 (–4.48 to 5.18)	1.70 (–3.33 to 6.72)	14.43 (–0.84 to 29.70)	**6.86 (0.58 to 13.14)**
1 month vs 3 months	0.31 (–2.59 to 3.20)	–1.83 (–5.19 to 1.54)	**15.91 (4.52 to 27.30)**	**12.07 (6.42 to 17.73)**
Control				
Baseline vs 1 month	0.47 (–3.84 to 4.77)	0.25 (–3.52 to 4.00)	5.68 (–4.27 to 15.63)	2.85 (–3.18 to 8.89)
Baseline vs 3 months	–1.79 (–5.97 to 2.38)	–2.53 (–5.30 to 0.24)	**16.14 (3.19 to 29.09)**	**7.69 (1.16 to 14.23)**
1 month vs 3 months	–2.01 (–6.45 to 2.42)	**–2.65 (–4.78 to –0.53)**	10.86 (–1.23 to 22.95)	5.19 (–2.04 to 12.43)

Data are presented as mean %Δ (95% confidence interval). Change scores were calculated as the later time point minus the earlier time point normalized to the earlier time point (e.g., %Δ = 100 × [(1 month – baseline)/baseline]). Values represent the mean %Δ and the upper and lower bounds of the 95% confidence interval of the change. Change scores were considered statistically significant (**bold text**) if the 95% confidence interval did not include 0.

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The time x limb interaction effects were significant for the EI of the RF ($P < 0.001$, Fig. 2) and VL ($P < 0.001$, Fig. 3) in the ACLR group. The EI values of the RF and VL were similar in both limbs at baseline ($P = 0.784$ and $P = 0.145$, respectively) but were greater in the ACLR limb compared with the contralateral limb at 1 month ($P = 0.003$ and $P < 0.001$, respectively) and 3 months ($P = 0.002$ and $P < 0.001$, respectively).

Secondary analysis of the change scores between successive time points indicated that the increases in EI in the ACLR limb from baseline to 1 month and from baseline to 3 months were significant for both RF and VL, but the associated changes from 1 to 3 months were not significant (Table 2).

Quadriceps CSA. The time x group interaction effects were significant for CSA of the RF ($P < 0.001$, Fig. 4) and VL ($P < 0.001$, Fig. 5). RF CSA in the ACLR limb was smaller at 1 month compared with baseline ($P = 0.001$) and 3 months ($P = 0.002$) but was not different from baseline at 3 months ($P = 0.140$). VL CSA in the ACLR limb was smaller at 1 month ($P < 0.001$) and 3 months ($P < 0.001$) compared with baseline but was greater at 3 months compared with 1 month ($P < 0.001$). There were no differences in CSA across time in the control group for either muscle ($P = 0.015$–0.420).

RF CSA was similar between the groups at baseline ($P = 0.951$) but was smaller in the ACLR limb compared to controls at 1 month ($P = 0.011$) and 3 months ($P = 0.013$), although these differences failed to reach statistical significance due to the conservative nature of the Bonferroni correction (adjusted pairwise $\alpha = 0.006$). VL CSA was similar between the groups at baseline ($P = 0.221$) but was smaller in the ACLR limb at 1 month ($P < 0.001$) and 3 months ($P = 0.003$) compared with the control group. The MCID values for RF and VL CSA were 0.55 and 1.09 cm^2, respectively, at 1 month and 0.63 and 1.36 cm^2, respectively, at 3 months. At both 1 and 3 months, the differences in the CSA of the RF and VL between the ACLR limb and the control group exceeded the MCID.

The time x limb interaction effects were significant for RF CSA ($P = 0.017$, Fig. 4) and VL CSA ($P < 0.001$, Fig. 5) in the ACLR group. RF CSA was smaller in the ACLR limb

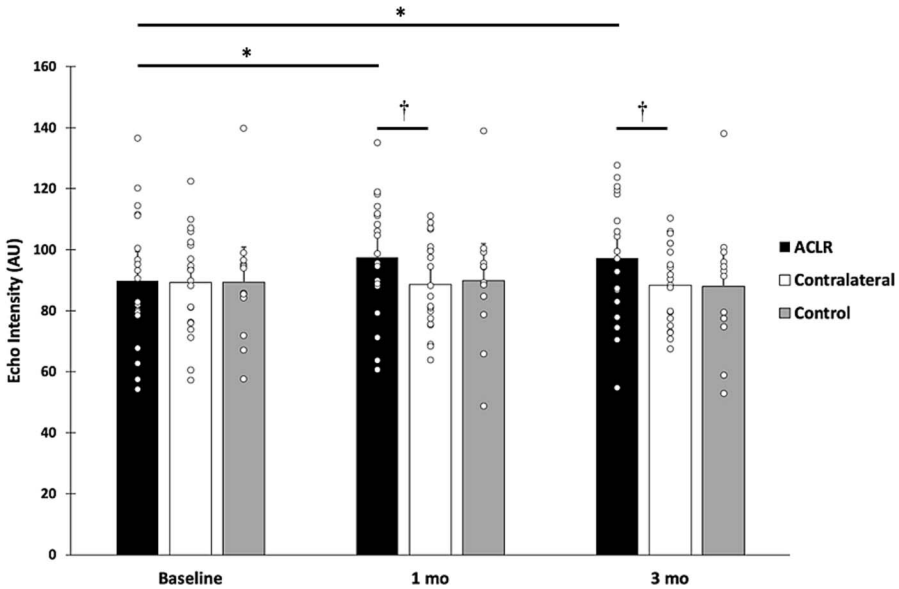

FIGURE 2—RF EI in the ACLR limb was significantly greater at 1 and 3 months post-ACLR compared to baseline and the contralateral limb, indicating declines in muscle quality that remained present at 3 months. The ACLR limb was not significantly different from control at any time point, but these group differences exceed the MCID at 1 month (2.16) and 3 months (2.21). *Open circles* represent individual subject means. *Error bars* represent the upper bound of the 95% confidence interval of the limb mean. * Significantly different versus baseline. † Significantly different versus contralateral limb.

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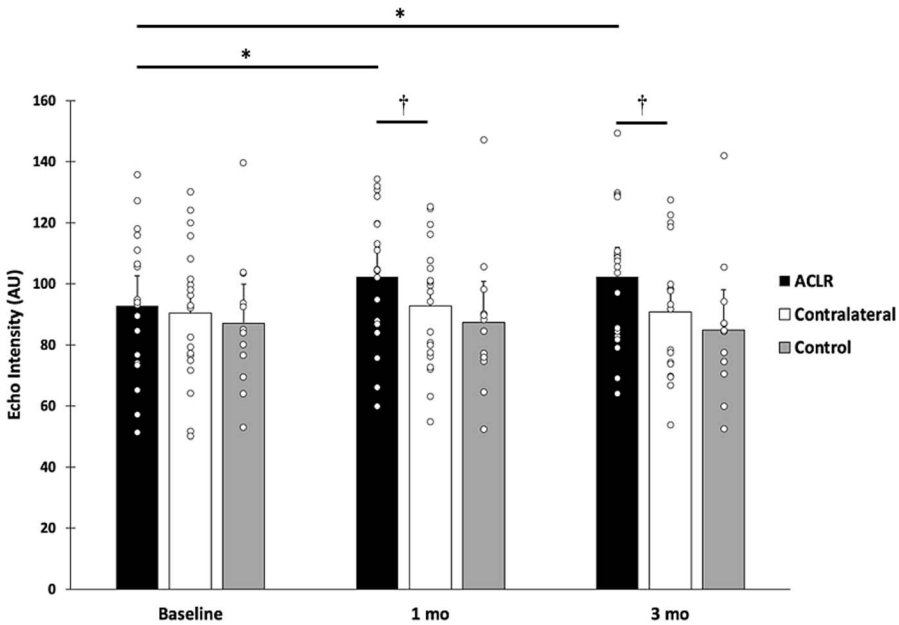

FIGURE 3—VL EI in the ACLR limb was significantly greater at 1 and 3 months post-ACLR compared to baseline and the contralateral limb, indicating declines in muscle quality that remained present at 3 months. The ACLR limb was not significantly different from control at any time point, but these group differences exceed the MCID at 1 month (1.86) and 3 months (1.78). *Open circles* represent individual subject means. *Error bars* represent the upper bound of the 95% confidence interval of the limb mean. * Significantly different versus baseline. † Significantly different versus contralateral limb.

compared with the contralateral limb at 1 month ($P < 0.001$) and 3 months ($P < 0.001$) but not at baseline ($P = 0.335$). VL CSA was smaller in the ACLR limb compared with the contralateral limb at baseline ($P < 0.001$), 1 month ($P < 0.001$), and 3 months ($P < 0.001$).

Secondary analysis of the change scores between successive time points indicated that the decreases in CSA in the ACLR limb from baseline to 1 month were significant for both the RF and the VL. Similarly, the increases in CSA from 1 to 3 months were significant for both the RF and the VL such that RF CSA was no longer different from baseline, although VL CSA was still significantly smaller compared to baseline.

Associations between changes in CSA and EI. A larger decrease in CSA from baseline to 1 month in the ACLR limb was associated with a larger increase in EI over the same time interval for the VL ($r = -0.584$, $P = 0.007$), but the

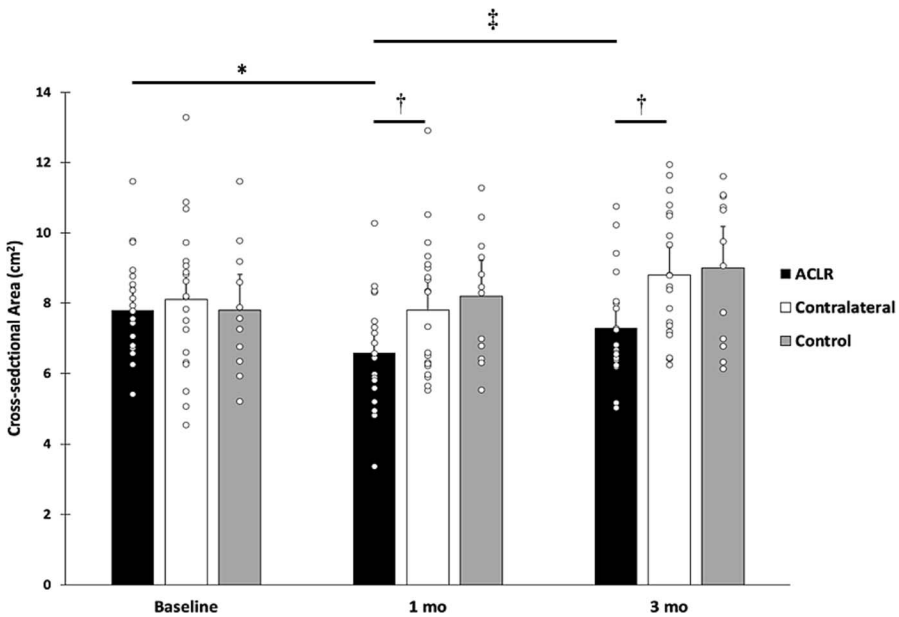

FIGURE 4—RF CSA in the ACLR limb was significantly smaller at 1 month post-ACLR compared with baseline and 3 months post-ACLR and was smaller than the contralateral limb at 1 and 3 months post-ACLR. The ACLR limb was not significantly different from control at any time point, but these group differences exceed the MCID at 1 month (0.55) and 3 months (1.36). *Open circles* represent individual subject means. Error bars represent the upper bound of the 95% confidence interval of the limb mean. * Significantly different versus baseline. † Significantly different versus contralateral limb. ‡ Significantly different versus 3 months.

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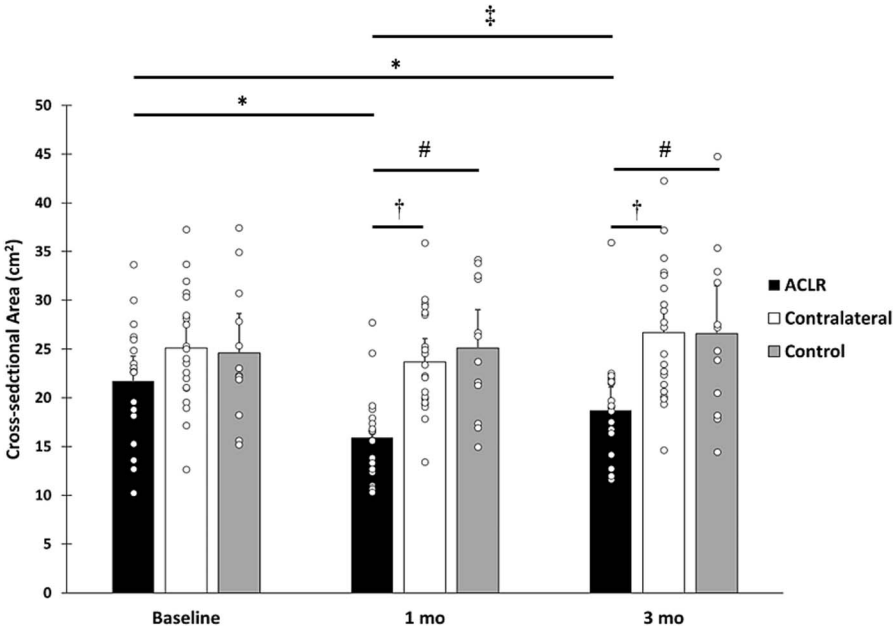

FIGURE 5—VL CSA in the ACLR limb was significantly smaller at 1 and 3 months post-ACLR compared with baseline and was significantly larger at 3 months compared with 1 month. CSA was smaller in the ACLR limb than the contralateral limb at 1 and 3 months post-ACLR. The ACLR limb was not significantly different from control at any time point, but these group differences exceed the MCID at 1 month (0.55) and 3 months (1.36). *Open circles* represent individual subject means. Error bars represent the upper bound of the 95% confidence interval of the limb mean. *Significantly different versus baseline. †Significantly different versus contralateral limb. ‡Significantly different versus 3 months. #Significantly different versus control.

changes in EI and CSA over this interval in the RF were not correlated ($r = -0.236$, $P = 0.316$) (Fig. 6). Similarly, the change in CSA from 1 to 3 months was not associated with the change in EI over the same time period for the RF ($r = -0.312$, $P = 0.181$) or VL ($r = -0.029$, $P = 0.903$).

DISCUSSION

We evaluated QMQ over the first 3 months post-ACLR using EI obtained from ultrasound imaging. The results indicate that QMQ declined over the first month in the ACLR limb and did not improve by 3 months. Quadriceps atrophy also occurred over the first month after ACLR, but CSA increased from 1 to 3 months. Although the changes in CSA and EI were correlated over the first month post-ACLR, these same changes from 1 to 3 months were not correlated. As such, QMQ and CSA appear to change independently after ACLR and may uniquely influence quadriceps function.

Both the RF and the VL in the ACLR limb displayed greater EI at 1 and 3 months compared with baseline and compared

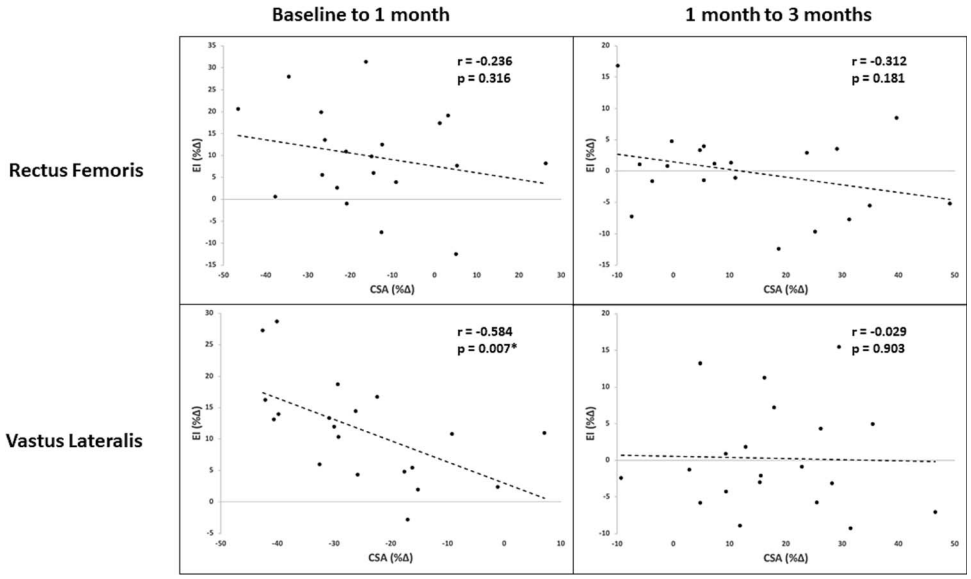

FIGURE 6—A larger decrease in CSA of the VL was associated with a larger increase in EI in the ACLR limb from baseline to 1 month post-ACLR. Changes in CSA and EI over this interval were not associated in the RF, and these changes from 1 the 3 months post-ACLR were not associated in either muscle.

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with the contralateral limb, indicating declines in QMQ. Additionally, although not statistically significant, the differences in EI between the ACLR and the control limbs 1 and 3 months post-ACLR exceed the MCID. These findings are contrary to Garcia et al. (33), who reported no differences in QMQ 9 wk post-ACLR or at the time of return sport (~41 wk post-ACLR) compared with baseline. However, this study used used ultrasound imaging to estimate %fat within the muscle rather than EI, potentially explaining discrepancies in the findings. Our findings agree with a second study by Garcia et al. (32), which reported poorer QMQ identified via ultrasound EI in the ACLR limb compared with the contralateral limb ~4 yr post-ACLR, and a study by Lindström et al. (31), which reported similar findings with CT imaging. This suggests that QMQ declines within the first month after ACLR, is not mitigated with rehabilitation, and persists for several years after surgery.

Poorer QMQ is indicative of greater noncontractile tissue concentration within the muscle (18,39,40) and is linked to deficits in muscle activation and functional limitations (20–22). In individuals with ACLR and knee osteoarthritis, poorer QMQ is associated with poorer strength, function, patient-reported outcomes, and quality of life (23,32–35). As such, poorer QMQ may hinder recovery after ACLR and contribute to chronic quadriceps dysfunction. Quadriceps dysfunction has been linked to aberrant gait biomechanics that contribute to the development of posttraumatic knee osteoarthritis (7–9). As such, future longitudinal research is needed to evaluate the long-term influence of QMQ on quadriceps function, gait biomechanics, and knee joint health after ACLR.

Our results also indicate a rapid reduction in quadriceps CSA after ACLR in agreement with previous literature (11,12,15). Both the VL and the RF were smaller at 1 month compared with baseline in our ACLR cohort, but the CSA of both muscles increased from 1 to 3 months, suggesting that rehabilitation was effective at inducing muscle hypertrophy in agreement with previous reports (11,41). By contrast, the EI of both RF and VL increased relative to baseline at 1 month and remained elevated at 3 months. In fact, the CSA of VL and RF increased by 17.9% and 13.4%, respectively, from 1 to 3 months, whereas the corresponding changes in EI were 0.3% and −0.2%. Furthermore, changes in CSA over each time interval were not associated with the change in EI. These data suggest that CSA and EI change independently after ACLR and that current rehabilitation approaches likely do not address early declines in QMQ. Additionally, CSA of the VL was already smaller in the ACLR limb compared with the contralateral limb at baseline, whereas EI did not differ between limbs. This finding differs from previous investigations reporting poor QMQ assessed via fibrotic tissue content after ACL injury but before ACLR (13,17). This discrepancy may suggest that ultrasound EI is more sensitive to the fatty infiltration of the muscle than to fibrotic tissue formation (40) or is influenced by ACLR rather than the initial ACL injury, but future research is necessary to evaluate these notions.

Stockmar et al. (45) reported that atrophy and inhibition of the quadriceps cause a metabolic shift from greater reliance on glycolytic activity to more oxidative utilization for energy production and greater dependence on fat as an energy source. These authors theorized that the quadriceps engages in sustained low-level compensatory activity after ACLR to promote knee stability that eventually leads to an increase in Type I muscle fiber concentration and a decrease in Type II fiber concentration. Noehren et al. (13) reported a reduction in the size of Type II muscle fibers that was accompanied by a decrease in the number of satellite cells per muscle fiber and an increase the extracellular matrix volume. Lopresti et al. (46) also reported a reduction in Type II muscle fiber size in the ACLR limb compared with the contralateral limb, yet the circumference of the limbs remained equal. Therefore, it is possible that increases in muscle CSA after early atrophy may be influenced by fatty infiltration to support a greater reliance on oxidative Type I fibers and formation of fibrous tissue (i.e., poor QMQ). However, our data indicate that although quadriceps CSA increased from 1 to 3 months post-ACLR, EI essentially remained constant over this interval, and the changes in CSA and EI were not correlated, suggesting that these morphological changes occur independently. As such, the changes in CSA were likely driven by hypertrophy. Tracking QMQ throughout rehabilitation may provide clinicians an objective indication of patient progress that measures of muscle size do not afford.

The results of this study should be considered within the context of its limitations. Our sample sizes were relatively small, and some statistically nonsignificant findings were underpowered but may be clinically relevant as indicated by comparisons to MCID. Although EI has been associated with fatty and fibrous tissue concentrations in skeletal muscle (18,39,40), it can be influenced by a number of factors not indicative of muscle function (e.g., hydration status, ultrasound probe orientation), and the extent to which it reflects fatty infiltration versus fibrous tissue formation is unclear (47). Additionally, we only evaluated VL and RF and did not consider potential changes of the vastus intermedius or medialis because of the inconsistent image clarity of the deep borders of these muscles. Images were obtained at 50% of the length of the thigh and may not account for more proximal or distal muscular characteristics. We did not account for the potential influence of ACLR graft type; however, recent literature indicates similar quadriceps size and strength in individuals with patellar tendon and quadriceps tendon grafts (48), the two most common graft types in our sample. We did not account for the influence of sex, but none of the change scores reported in Table 2 differed across sex, suggesting it did not influence our findings. All individuals underwent surgery from one of three orthopedic surgeons from the same clinic; however, rehabilitation was not standardized among the participants. Physical activity and exercise for the control group were not accounted for and could have influenced changes in muscle size or quality. CSA of both VL and RF slightly increased from baseline to 3 months in the control group and may be attributed to exercise or training regimens not accounted for in the study protocol. Lastly, we did not account for the time between ACL injury and

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ACLR, between baseline study visits and ACLR, or between study visits at 1 and 3 months post-ACLR. However, we evaluated simple Pearson correlations between each change score in Table 2 and the number of days in each of the aforementioned time intervals, separately. Of these 36 correlations, only 2 were significant (greater days pre-ACLR were associated with a greater increase in RF EI of the contralateral limb from baseline to 1 month [$r = 0.450$, $P = 0.047$] and from baseline to 3 months [$r = 0.505$, $P = 0.023$]). These data suggest that the covariates had little-to-no influence on our findings.

CONCLUSIONS

The results of the current investigation suggest that QMQ declines rapidly after ACLR and does not improve within the first three postoperative months. These changes in QMQ are accompanied by a decrease in CSA, yet increases in muscle size from 1 to 3 months occur in the absence of improvements in muscle quality. Deficits in QMQ are linked to deleterious effects on quadriceps dysfunction and functional ability and may contribute to poor long-term outcomes. Future research is necessary to determine how QMQ changes across the course of rehabilitation and whether these changes influence long-term knee joint health. Additionally, various forms of exercise have been demonstrated to improve QMQ in the elderly and individuals with knee osteoarthritis (49,50); thus, future research is necessary to identify approaches for reversing changes in quadriceps composition after ACLR.

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